New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: _____ SIGNATURE: _____

ID NUMBER: _____ EXAM ROOM: _____ GROUP: _____

Midterm Exam

COURSE NAME: Computer Vision COURSE NUMBER: CS418

EXAMINATION DURATION: 120 MINUTES EXAM VERSION: A

ADDITIONAL MATERIALS ALLOWED TO USE: Pens and a non-programmable scientific calculator only.

Do not open this exam paper until you are told to begin.

READ THESE INSTRUCTIONS FIRST. Breaking the rules can lead to a loss of points.

- Coverage: Weeks 1–5. Answer all 26 questions: 20 multiple-choice questions (MCQs) and 6 written questions in 5 problems. Total: 100 points. Use the spaces provided. If you continue on the back, write the question number there and note where your answer continues.
- Each question has its own data. Do not use data from another question. Each MCQ is worth 2 points. Wrong answers receive 0; no points are deducted. In written answers, show calculations and explain your reasons using the case given. Correct steps can earn points.
- Sign your name and write your student ID on every page.
- Follow the University Examination Rules. This is a closed-book exam. Do not use books, notes, laptops, phones, smartwatches, the internet or AI tools. Do not share calculators or other materials.

FOR INSTRUCTOR USE ONLY. STUDENTS MUST NOT WRITE IN THIS TABLE.

Problem	Type of Questions	Points	Score
1	Multiple Choice — Questions 1–20	40	
2	Short Answer — Questions 21–22	20	
3	Problem Solving — Questions 23–24	20	
4	Analytical Question — Question 25	10	
5	Open Question — Question 26	10	
Total		**100**	

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Problem 1 Multiple Choice (40 points)

Questions 1–5. Circle one answer (A–E) for each question. Each question is worth 2 points.

1. [2 points] A parcel is a package or box. Its printed label moves by one pixel in the image, but the printing does not change. Why can a raw pixel-by-pixel comparison still give a large difference?

A. It now compares pixels from different parts of the printed pattern
B. A one-pixel shift always changes the true label class
C. A shift adds a new colour channel to the image
D. Every pixel value must stay at its original position after a shift
E. None of the above

2. [2 points] An image of a printed label has shape (24, 40, 3). Its values are stored as I[row, column, channel]. Indexing starts at 0: the first row, column and channel each have index 0. What is wrong with I[30, 10, 1]?

A. Channel index 1 is outside the image
B. Row index 30 is outside the image, but column index 10 is valid
C. Column index 10 is outside the image
D. There is no error because 30 is less than 40
E. None of the above

3. [2 points] A camera stays in the same position. Only the lamp brightness changes, and the recorded pixel values change. Which conclusion is supported?

A. The parcel must now have a rip
B. The image must now have an extra channel
C. Different pixel values alone do not prove that the parcel shape changed
D. The camera must have moved
E. None of the above

4. [2 points] Two images of a printed parcel label have the same intensity histogram. One image was made by moving the pixels to different positions without changing their values. What must remain the same?

A. The positions of the printed lines
B. The number of pixels at each intensity level
C. The direction of the printed text
D. The positions of all boundaries between intensity levels
E. None of the above

5. [2 points] To brighten an 8-bit image, add 20 to each pixel value. Then clip: replace any value above 255 with 255. Original values 240 and 250 both become 255. What does this show?

A. Subtracting 20 always gives the original values
B. The output does not always tell us the exact original intensity
C. Clipping keeps every difference between bright pixels
D. The output value must become zero instead
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 6–10. Circle one answer (A–E) for each question. Each question is worth 2 points.

6. [2 points] A known neutral grey patch is recorded as RGB [180, 150, 90]. Multiply the R, G and B values by [1, 1.2, 2], with no clipping. Which interpretation is correct?

A. The corrected values are [180, 151.2, 92]
B. The correction changes the physical material of the patch
C. One reference correction is guaranteed to work under every lighting condition
D. The values become [180, 180, 180], balancing the reference without changing its material
E. None of the above

7. [2 points] A large gradient is found at a printed line and at the edge of a rip in the label. Why is the gradient alone not enough to decide which one is damage?

A. Both can produce a large local change in intensity
B. A gradient measures only the image height and width
C. Printed lines always have zero gradient
D. Every nonzero gradient is a real rip
E. None of the above

8. [2 points] At the centre of the pixel row [2, 2, 8], cross-correlation with kernel [−1, 0, 1] gives 6. What result does mathematical convolution give at the same position, using the same kernel?

A. 6, because reversing any kernel leaves it unchanged
B. 12, because convolution doubles the response
C. −6, because convolution reverses this kernel
D. 0, because the centre weight of the kernel is zero
E. None of the above

9. [2 points] A strong blur removes camera noise, but it also makes a thin rip disappear from the image. Which conclusion is best supported?

A. The strongest blur is always the best preprocessing
B. Every smoothing filter must keep all thin boundaries
C. Smoothing should reduce noise while keeping details needed for the task
D. Removing the noise proves that the detected rip was not real
E. None of the above

10. [2 points] Two different linear kernels each have weights that sum to 1. They give the same result on a constant patch, where every pixel has the same value. Must they also give the same result on every other patch?

A. No; different weights can give different responses to changes across the patch
B. Yes; the sum of the weights determines the result for every image
C. Yes; normalization makes all kernels identical
D. No; a normalized kernel cannot process a constant patch
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 11–15. Circle one answer (A–E) for each question. Each question is worth 2 points.

11. [2 points] Intensity changes strongly across the width of a printed stripe, but changes very little along its length. Why can this stripe be a poor place to locate a keypoint precisely?

A. The whole region must have no intensity change
B. Every stripe is a strong corner in two directions
C. Moving along the stripe can leave the nearby image almost unchanged
D. A larger gradient guarantees one unique position along the stripe
E. None of the above

12. [2 points] A descriptor has nearest distance 4 and second-nearest distance 5. The rule is: accept only if nearest distance divided by second-nearest distance is less than 0.7. Which decision follows?

A. Accept: the nearest distance is smaller than the second-nearest distance
B. Reject: 4/5 = 0.8 does not pass the stated test
C. Accept: the nearest neighbour is always the correct parcel
D. Reject: a distance of 4 always proves the match is wrong
E. None of the above

13. [2 points] A printed parcel label rotates in the image. A descriptor measures gradient directions relative to the dominant local orientation. How can this help?

A. It makes the descriptor less sensitive to rotation in the image plane
B. It guarantees a correct match for every repeated pattern
C. It makes a blurred patch identical to a sharp patch
D. It removes the need to find the keypoint position
E. None of the above

14. [2 points] A known camera movement should move a label corner from x = 20 to x = 30. The candidate with the best descriptor match is at x = 45. What should be checked?

A. Only whether this candidate has the smallest descriptor distance
B. Only whether the two images have the same width
C. Whether the candidate has the largest x coordinate
D. Whether the candidate position agrees with the known camera movement
E. None of the above

15. [2 points] A three-class parcel-label classifier gives probabilities [0.44, 0.34, 0.22]. It predicts the first class. Is this decision necessarily wrong because 0.44 is below 0.50?

A. Yes; a predicted class must always have probability above 0.50
B. Yes; all three classes must be predicted together
C. Yes; these probabilities cannot sum to 1
D. No; 0.44 is the largest value, but the true class still needs separate evidence
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 16–20. Circle one answer (A–E) for each question. Each question is worth 2 points.

16. [2 points] Before applying softmax, a program subtracts the same value, 5, from every class score. In exact arithmetic, what changes?

A. Neither the class probabilities nor their order changes
B. Only the predicted class changes
C. All probabilities become negative
D. The class with the smallest original score becomes the winner
E. None of the above

17. [2 points] A classifier is correct on all 180 undamaged parcels but wrong on all 20 parcels with torn labels (labels with a rip). Which report best describes this result?

A. The 90% overall accuracy proves good performance on both groups
B. The 90% overall accuracy hides 0% accuracy on torn-label parcels
C. Twenty wrong answers mean the overall accuracy is 20%
D. Correct answers for undamaged parcels prove correct answers for damaged parcels
E. None of the above

18. [2 points] Two models both correctly classify one parcel as damaged. They assign probabilities 0.55 and 0.85 to this true class. Using loss = −ln(p_true), which statement is correct?

A. The losses must be equal because the predicted classes are equal
B. The model that assigns 0.55 has the lower loss
C. Both losses must be zero because both predicted classes are correct
D. The model that assigns 0.85 has the lower loss, although both predictions are correct
E. None of the above

19. [2 points] A classifier is correct on one parcel image. After a one-pixel shift, it is wrong on that image. What does this experiment establish?

A. Every possible image shift will cause a wrong prediction
B. The original prediction must also have been wrong
C. A one-pixel shift cannot change the classifier input
D. This pair shows shift sensitivity, but does not measure performance under all shifts
E. None of the above

20. [2 points] A team tries several learning rates. Each time, it uses test results to choose the best rate. Which change best keeps a separate, independent final evaluation?

A. Rename the repeatedly used test set as untouched
B. Choose the rate using only the lowest training loss
C. Choose the rate using validation data and keep test data for the final evaluation
D. Add the test labels to the training labels after each trial
E. None of the above

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Problem 2 Short Answer (20 points)

Question 21 Check a warehouse camera output [10 points]

A warehouse camera takes RGB images with height 30 pixels and width 50 pixels. A parcel means a package or box. Store image values as I[row, column, channel], with indexing starting at 0. Each channel value uses 8 bits, equal to 1 byte. Ignore compression and metadata when counting bytes. Part (b) uses separate small image patches.

(a) [3 points] A programmer creates an array with shape (50, 30, 3). Give the correct shape and the number of bytes needed. Explain why the same number of bytes does not make the proposed shape correct.

(b) [3 points] Two stored label templates are A = [[0, 10], [10, 0]] and B = [[4, 14], [14, 4]]. A brighter photo of label A is Q = [[5, 15], [15, 5]]. Use MAD: find the absolute difference at each matching position, then average the four differences. Calculate MAD from Q to each template. Which template wins? Explain why this comparison can choose the wrong label here.

(c) [4 points] Two photos of an undamaged parcel have different pixel values. Plan a controlled comparison to test whether lamp brightness explains the difference. State:
1. What you would change.
2. What you would keep the same.
3. What the comparison could show.

Answer and working

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Problem 2 Short Answer continued (20 points)

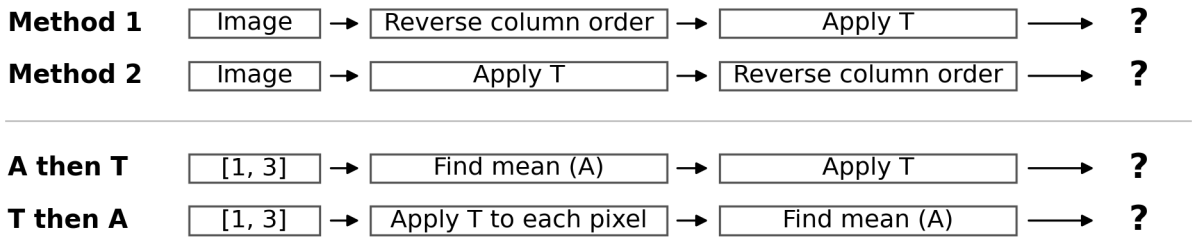
Question 22 Check the order of image operations [10 points]

The matrix shows a grayscale label image. Its pixel values are 0, 1, 2 and 3. To form J, apply T(x) = min(3, 2x) to each pixel: multiply by 2, then replace any result above 3 with 3 (clipping). Histograms here are pixel counts, not percentages.

Column 1	Column 2	Column 3	Column 4
0	0	3	3
0	0	3	3
1	1	2	2
1	1	2	2

(a) [3 points] Give the histogram before T and the histogram after T. List the counts in level order [0, 1, 2, 3]. Also show how T changes each of the four possible input levels.

(b) [3 points] Can an operator always find the exact original value of every pixel using only J? Evaluate this claim. Use two original intensity values that give the same output, and explain what information is lost.



(c) [4 points] Use the diagram to compare these orders.
1. Must Method 1 and Method 2 give the same final image? Explain. Reversing columns means reading each row from right to left.
2. For [1, 3], A replaces the pair with its arithmetic mean. Use real numbers without rounding. Find A then T and T then A. Explain any difference.

Answer and working

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Problem 3 Problem Solving (20 points)

Question 23 Check image changes and possible damage [10 points]

The small image patch below has two intensity levels and comes from a printed parcel label. Use the given Kx and Ky kernels to find the centre responses Gx and Gy. Use cross-correlation: multiply matching entries and add the products, without flipping the kernels. Only the centre response is needed, so no padding is used. Use real-valued intensities without clipping. These are derivative kernels, not averaging kernels.

Column 1	Column 2	Column 3
2	2	8
2	2	8
2	2	8

Kernel	Row 1	Row 2	Row 3
Kx	−1, 0, 1	−1, 0, 1	−1, 0, 1
Ky	−1, −1, −1	0, 0, 0	1, 1, 1

$$|G| = \sqrt{G_x^2 + G_y^2}$$

(a) [3 points] Calculate Gx, Gy and the gradient magnitude. Show enough working to make the positive or negative signs clear.

(b) [4 points] Start again from the original patch for each of these two separate cases:
Case 1: Add 5 to every pixel value.
Case 2: Multiply every pixel value by 2.
Predict the new Gx and Gy in each case without repeating all products. Explain your predictions using the kernel weights. These changes are made to the pixel values, not directly to Gx and Gy.

(c) [3 points] An operator says, "The gradient is not zero, so the label must have a rip." Is this conclusion justified? Explain. Suggest one more comparison that could help distinguish a normal printed boundary from damage.

Answer and working

Problem 3 Problem Solving continued (20 points)

Question 24 Check a corner and a proposed match [10 points]

The first table gives structure matrices for two regions of a printed label. Use the Harris formula with k = 0.04. You do not need to calculate eigenvalues. The second table gives descriptor distances for queries P and Q. Accept a match only if nearest distance / second-nearest distance is strictly less than 0.7.

Region	Structure matrix
A	[[6, 0], [0, 6]]
B	[[30, 0], [0, 0]]

Query	Nearest distance	Second nearest distance
P	9	10
Q	14	28

For a symmetric matrix M = [[a, b], [b, c]], use these formulas:

$$R = \det(M) - k[\text{trace}(M)]^2$$

$$\det(M) = ac - b^2, \qquad \text{trace}(M) = a + c$$

(a) [4 points] Calculate the Harris response for A and B. Choose the better corner candidate. Explain why the larger single diagonal value in B is not enough to make B the better corner.

(b) [3 points] This is a separate change: multiply every image-gradient value used to build M by 2. The new structure matrix is therefore 4M. Does the new Harris response equal 4R or 16R? Explain using the determinant and the squared trace. Also state whether a positive, negative or zero response changes sign.

(c) [3 points] Apply the ratio test to P and Q. A colleague says that passing this test proves the match is to the correct printed label in the other image. Is the colleague's claim justified? Explain your answer, and name one more consistency check that does not rely only on the ratio.

Answer and working

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Problem 4 Analytical Question (10 points)

Question 25 Check a linear classifier [10 points]

A classifier reads LEFT or RIGHT arrows on parcel labels. Parts (a) and (b) use the feature vector x = [1, 3] and the weights in the table. A class score is a weighted sum plus its bias. Predict the class with the larger score. Softmax converts scores to probabilities. Part (c) gives separate predictions for another labelled example.

Class	Weight for x1	Weight for x2	Bias
LEFT	2	0	−1
RIGHT	0	1	−1

$$z_k = w_{k1} x_1 + w_{k2} x_2 + b_k$$

$$p_k = \frac{e^{z_k}}{e^{z_{\text{LEFT}}} + e^{z_{\text{RIGHT}}}}$$

$$L = -\ln(p_{true})$$

Here k is LEFT or RIGHT; p_true is the probability of the true class.

(a) [4 points] Calculate the LEFT and RIGHT scores and give the predicted class. A student treats these scores as probabilities. Explain why that interpretation is not valid.

(b) [3 points] A program adds 7 to both class scores before applying softmax. Do the probabilities or predicted class change? Explain using the supplied softmax formula, without calculating large exponentials.

(c) [3 points] In a separate example, the true class is LEFT. Training changes the probabilities [LEFT, RIGHT] from [0.24, 0.76] to [0.46, 0.54]. A student says, "Nothing improved because the predicted class stayed the same." Is the student's claim correct? Use the predicted class and cross-entropy loss in your explanation. You do not need to calculate logarithms.

Answer and working

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Problem 5 Open Question (10 points)

Question 26 Choose a model for reading arrow labels [10 points]

A model reads arrows printed on parcel labels. Its classes are LEFT and RIGHT, according to the direction of the arrow. The table shows losses at four training checkpoints (saved versions of the model). The training and validation sets stay fixed. A separate test set has not been used. For part (b), use the given update formula; its gradient is separate from the data in the table.

Epoch	Training loss	Validation loss
1	0.80	0.90
3	0.50	0.55
6	0.30	0.42
9	0.15	0.60

$$w_{\text{new}} = w_{\text{old}} - \eta g$$

(a) [4 points] A researcher chooses epoch 9 because it has the lowest training loss. Choose a checkpoint using the evidence in the table. Explain what happens in the later epochs. State how the separate test set should be used.

(b) [2 points] One weight is −0.20. Its gradient is +0.50, and the learning rate is 0.10. Calculate the next weight. Explain why a positive gradient does not mean that the weight should increase.

(c) [4 points] This is a separate one-weight example. Its loss is $L(w) = w^2/2$ and its gradient is $g = w$. Start at $w = 2$ for each trial. Use learning rate 1.5 in Trial A and 2.5 in Trial B. Calculate the new weight and loss in each trial. Both new weights are negative. Does that sign change alone prove that the loss increased? Explain using your results.

Answer and working

END OF EXAM